

Syeda Mariah Banu

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MACHINE LEARNING ENGINEER — Computer Vision · LLM/Agentic · FastAPI/Kubernetes · MLflow · AWS

- Ships production Computer Vision and LLM/agentic systems; designs reliable training/serving pipelines, evaluation harnesses, and autoscaled inference on Kubernetes/EKS.
- Partners with product/SRE on SLOs, rollback safety, and observability; improves model quality and iteration speed while balancing cost/latency tradeoffs.
- Builds FastAPI services, MLflow registries, and request shaping for near real-time latency and high throughput; drives guardrails, drift detection, and incident response quality.

CORE SKILLS

Languages: Python, SQL, (Java), Bash

ML/DL: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, Hugging Face

MLOps/Platforms: MLflow, Airflow, Kubeflow, SageMaker, Vertex AI, Azure ML, Weights & Biases

Serving/Systems: FastAPI, Docker, Kubernetes/EKS, KServe/Seldon, BentoML, ONNX/TensorRT, gRPC

Data/Compute: Pandas/NumPy, Spark/Ray, Parquet, Feature Store (Feast), Data Quality (Great Expectations)

LLM/RAG: Transformers, PEFT/LoRA, retrieval pipelines, vector DBs (FAISS/Pinecone),

LangChain/LlamaIndex, prompt/eval harnesses

Experimentation & Eval: offline/online eval, A/B testing, guardrails, drift detection (Evidently)

Observability & Ops: OpenTelemetry, Prometheus/Grafana, Datadog, Splunk; incident response/on-call

DevOps/Cloud: AWS (EKS, S3, Lambda, DynamoDB), GCP, Azure, Terraform/Helm, Argo CD, CI/CD (GitHub Actions/Jenkins)

EXPERIENCE

Comfortliv LLC — Machine Learning Engineer • May 2025 – Present

- Delivered end-to-end CV classification pipeline (data curation → PyTorch training → FastAPI inference on EKS) with significant accuracy and recall improvement in production.
- Built LLM/agentic workflows (prompting, tool-calling, RAG) for data extraction, annotation/labeling, and report generation; dramatically reduced manual effort and sped turnaround.
- Productionized inference with autoscaling and request shaping for near real-time latency and high-throughput workloads.
- Implemented MLflow experiments and model registry to substantially shorten iteration and deployment time.
- Authored deployment runbooks/checklists that improve stability and reduce change-related issues.

Cedar Bluff Technologies — Computer Vision Lead • Jul 2024 – May 2025

- Led CV pipeline (annotation → training/fine-tuning → real-time inference); improved internal benchmarks from mAP@50 21.1% / Precision 17.8% / Recall 5.7% → mAP@50 99.2% / Precision 98.4% / Recall 96.7% on internal dataset.
- Operated EKS-hosted inference with feature-flagged releases and automated rollback; raised reliability and reduced incident risk.
- Delivered a stakeholder-facing product demo in 10 days, resulting in funding approval and a new client acquisition.
- Built dashboards and on-call runbooks that accelerated triage and improved overall service stability.

Southern Illinois University Edwardsville — Research Assistant (Plant Vision Lab) • Jan 2023 – May 2024

- Created a novel segmentation framework integrating deep learning techniques with correction factors to overcome estimation challenges.

- Improved yield estimation accuracy, achieving $R^2 = 0.96$ and reducing mean absolute error (MAE) by 10% compared to previous methods.
- Built reproducible training/evaluation harness with ablations and cross-validation; improved experimental clarity and decisions.
- Implemented lightweight APIs exposing datasets, predictions, and metrics to collaborators; significantly reduced time-to-insight.

Charter Communications (Consultant via Infosys) — Development & Automation Engineer • Aug 2023 – Dec 2023

- Automated 103 BDD cases (Cucumber, Selenium, REST Assured) in CI/CD; significantly reduced regression escapes and enabled rapid releases.
- Shipped Elasticsearch-based release-health and failure-triage dashboards; materially shortened detection-to-mitigation time.
- Implemented DynamoDB token-bucket rate limiting to protect downstream services during traffic spikes; improved resiliency.
- Integrated Okta (OIDC, SCIM, JIT) into test flows; reduced auth-related flakiness.

ServiceNow — Software Engineer • Jul 2020 – May 2021; Jul 2021 – Jul 2022

- Built Java/Spring Boot microservices for Virtual Agent on Kubernetes with Kafka; resiliency patterns improved responsiveness and service reliability.
- Standardized canary releases and rollback runbooks; deployment time decreased about 30% and release safety improved.
- Participated in on-call and authored knowledge-base articles; reduced repeat incidents through proactive documentation and fixes.

SELECTED PROJECTS

- CV Inference Service (Production): PyTorch → ONNX/TensorRT with FastAPI on EKS; request batching and autoscaling deliver near real-time p95 latency with high throughput; SLO dashboards and rollback playbooks improve operational safety.
- LLM Agent for Report Automation: Prompt toolkit with tool-calling and retrieval that substantially reduces analyst review time and improves output consistency; guardrails and evaluation harness.
- Data Prep & Evaluation Suite: Reproducible curation/augmentation pipelines plus ablation harnesses that materially shorten iteration cycles and improve experiment quality.
- RAG Knowledge Assistant: Retrieval-augmented QA over internal docs with guardrails and eval harness; noticeably improves answer relevance and reduces search time.

EDUCATION

M.S., Computer Science (CV/ML) — Southern Illinois University Edwardsville — GPA 4.0

CERTIFICATIONS

- IBM Data Science Professional Certificate
- AWS Solutions Architect – Associate (in progress)
- AWS Certified Machine Learning Engineer – Associate (in progress)

AWARDS & PUBLICATIONS

- MDPI: Journal of Imaging (Plant Phenotyping)
- Authorea: GCNet yield estimation
- NAPPN People's Choice — Top Fast Forward Speaker